

# Forest Change Detection using Sentinel-2 and Google Earth Engine

## 1. Objective

Evaluate whether Sentinel-2 vegetation indices can distinguish seasonal vegetation changes from actual forest loss over a selected region in Bavaria, Germany.

## 2. Study Area

- Region: Bavaria, Germany
- AOI defined using a rectangular bounding box.
- Time period analysed: January 2019 – December 2023
- Dataset: Sentinel-2 Level-2A (COPERNICUS/S2\_SR\_HARMONIZED)

## 3. Workflow

### 1) Data Preparation:

- a) Loaded Sentinel-2 Level-2A imagery.
- b) Filtered images to the Bavaria AOI and analysis period.
- c) Applied Scene Classification Layer (SCL) masking to retain:
  - Vegetation (SCL = 4)
  - Bare Soil (SCL = 5)
  - Water (SCL = 6)

### 2) Vegetation Indices:

Computed following three indices for every Sentinel-2 image:

- Normalized Difference Vegetation Index
- Enhanced Vegetation Index
- Bare Soil Index

### 3) Calculated Summer 2021 Composite

Generated a median composite for **June–August 2021** to obtain a representative summer image.

Displayed:

- RGB composite
- NDVI
- EVI
- BSI

#### 4) Observations recorded:

| Index | Intact Forest  | Agricultural Fields<br>(Pale Green) | Harvested / Ploughed<br>(Full Brown) |
|-------|----------------|-------------------------------------|--------------------------------------|
| NDVI  | 0.80 to 0.89   | 0.55 to 0.78                        | 0.2 to 0.35                          |
| EVI   | 0.42 to 0.70   | 0.40 to 0.72                        | 0.14 to 0.25                         |
| BSI   | -0.42 to -0.20 | -0.4 to -0.15                       | +0.02 to +0.14                       |

##### a) NDVI

- Intact forests consistently showed the highest NDVI values due to dense and healthy vegetation.
- Moderate NDVI values were observed in agricultural fields. depending on crop growth stage.
- Harvested or ploughed fields showed the lowest NDVI because vegetation cover had been removed which led to soil exposure.

##### b) EVI

- EVI followed a similar trend to NDVI but maintained better sensitivity in dense forests where NDVI was observed to saturate in a particular range.
- Actively growing crops produced EVI values comparable to forests during peak growth.
- Recently harvested fields showed substantially lower EVI values because of reduced vegetation biomass.

##### c) BSI

- Intact forests produced negative BSI values, indicating dense vegetation cover.
- Agricultural fields generally remained slightly negative while crops were present.
- Recently ploughed or harvested fields produced positive BSI values, indicating exposed soil.

\*Recently ploughed agricultural fields consistently exhibited positive BSI values, reflecting exposed bare soil, whereas intact forests maintained negative BSI values due to dense vegetation cover.

## 5) Stable Forest Analysis:

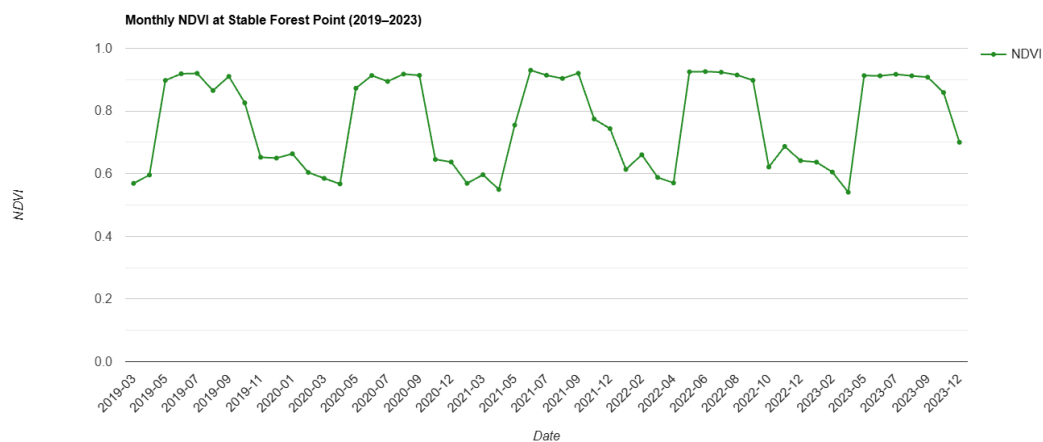
Selected a stable forest point with consistently high summer NDVI.

Extracted monthly NDVI median from January 2019 to December 2023.

### Point chosen:

[ Longitude: 11.736039805178699 Latitude: 47.93342334126962]

[NDVI (Summer 2021):  $\approx 0.91$ ]



*Monthly NDVI at Stable Forest Point (2019-2023)*

## Observations:

### A. Summer Peak NDVI

- Peak NDVI values consistently ranged between **0.90 and 0.93** during the summer growing season (June–August).
- These values indicate a dense, healthy and well-established forest canopy.

### B. Winter Trough NDVI

- Winter NDVI decreased to approximately **0.54 - 0.60**.
- The reduction reflects normal seasonal leaf-off conditions.
- Some winter months are absent because no valid Sentinel-2 observations remained after cloud and snow masking.

### **C. Seasonal Consistency (2019 - 2023)**

- A strong and repeatable annual phenological cycle was observed across all five years.
- NDVI increased during spring, peaked in summer, declined during winter and recovered again in the following growing season.
- This pattern remained highly consistent throughout the analysis period.

### **D. Year-to-Year Differences**

- Only minor variations in the timing and magnitude of the seasonal cycle were observed.
- No persistent downward trend in NDVI was detected.

\* Some winter months are absent because, after applying the Scene Classification Layer (SCL) cloud and snow mask, no valid Sentinel-2 observations remained for those months. These missing observations do not affect the overall seasonal pattern.

## **6) Hansen Forest Loss Comparison**

Loaded dataset: UMD/hansen/global\_forest\_change\_2023\_v1\_11

Selected a confirmed loss pixel satisfying:

- Tree Cover 2000 > 60%
- Loss Year  $\geq$  2020

Monthly NDVI was extracted using the same workflow.

Confirmed loss pixel considered:

- [Latitude: 11.45164, Longitude: 47.93448]
- treecover2000 = 97
- lossyear = 22



*Healthy Forest vs Confirmed Forest loss*

### ➤ **Observations:**

- The healthy forest exhibits a strong and repeatable seasonal NDVI cycle from **2019 - 2023**, with summer peaks of approximately **0.90 - 0.93** and winter values of **0.55 - 0.60**.
- The selected comparison pixel had **97% tree cover in 2000** and a confirmed Hansen **loss year of 2022**, which indicates a dense forest before disturbance.
- Prior to the recorded loss, the NDVI values of the loss pixel remained relatively high, consistent with forest vegetation.
- Around the confirmed loss year (**2022**), the NDVI decreased noticeably, reaching a minimum of approximately **0.37**.
- During the **2023** growing season, the NDVI recovered only partially and remained consistently below that of the healthy forest, suggesting that vegetation had not returned to its pre-disturbance canopy condition within the available observation period.

### ➤ **Required Observations**

- **Is the post-loss NDVI drop visually distinguishable from the normal winter trough?**

Yes. The healthy forest shows a temporary seasonal decline during winter followed by full recovery each summer. In contrast, the loss pixel exhibits a sustained reduction in NDVI after the confirmed loss year and does not recover to pre-loss levels during the following growing season.

- **How many growing seasons after the loss does the suppressed NDVI persist?**

Based on the available data, the reduced NDVI persists for at least one complete growing season (2023) following the confirmed loss in 2022. The observation period ends in 2023, so longer-term recovery cannot be assessed.

• **Is the drop magnitude larger or smaller than the normal seasonal swing?**

The magnitude of the NDVI decrease is comparable to the normal seasonal decline observed in the healthy forest. However, unlike the seasonal winter trough, which consistently recovers each year, the post-loss reduction remains suppressed during the subsequent growing season.

Normal seasonal drop observed:  $\approx 0.36$

Loss pixel drop observed:  $\approx 0.35$

**(Almost similar)**

➤ **Ambiguities / Notes**

- A relatively high NDVI value is observed for the loss pixel during late 2020 (November) despite the approaching winter season.
- Some winter months are absent because no cloud-free Sentinel-2 observations remained after applying the SCL cloud and snow mask.
- The Earth Engine chart displays a few x-axis labels out of chronological order due to the way ui.Chart renders grouped categorical dates. The underlying NDVI values and analysis are unaffected. (I tried to solve this issue but wasn't able to find a perfect solution)

\*GEE Code link: <https://code.earthengine.google.com/6d84a69276894de0f9be8d876cc32150>

**\*End of Report\***